

BATESVILLE 8 WNW, AR, USA

WMO: 722450

Lat: 35.8201N

Lon: 91.7812W

Elev: 139

StdP: 99.67

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-7.9	-16.6	0.9	-8.7	-13.9	1.1	-6.2	6.8	8.1	6.3	8.1	1.2	N/A	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	35.7	24.5	34.1	24.6	32.8	24.4	26.4	32.6	25.8	31.8	25.2	31.1	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.1	29.8	24.1	19.3	29.2	23.5	18.6	28.5	82.8	32.5	80.2	32.0	77.8	31.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	5.1	4.4	DB	-14.5	37.8	2.6	2.6	-16.3	39.7	-17.8	41.2	-19.2	42.7	-21.1	44.6	(4)
			WB	-15.3	27.4	2.7	0.6	-17.2	27.9	-18.8	28.2	-20.3	28.6	-22.2	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	2.5	4.7	10.0	14.4	19.5	24.6	25.9	25.4	21.8	15.2	8.8	4.5
	DBStd	9.45	5.79	6.11	5.64	4.65	3.95	2.62	2.71	2.83	3.46	4.74	5.20	5.15
	HDD10.0	775	242	170	72	13	1	0	0	0	0	9	82	186
	HDD18.3	2153	491	382	262	133	36	0	0	0	9	122	288	430
	CDD10.0	2539	9	22	71	146	297	437	492	479	356	170	47	14
	CDD18.3	875	0	1	3	15	74	187	234	221	114	24	2	1
	CDH23.3	8832	0	3	36	123	634	1956	2485	2228	1084	268	15	0
	CDH26.7	3828	0	0	2	9	159	862	1205	1081	441	69	1	0
Wind	WSAvg	1.7	2.2	2.3	2.3	2.1	1.5	1.4	1.2	1.2	1.2	1.4	1.8	1.9
Precipitation	PrecAvg	1249	89	84	118	126	116	88	103	77	109	104	120	115
	PrecMax	1650	176	171	298	290	232	151	168	175	241	332	302	209
	PrecMin	865	19	27	35	26	13	29	38	10	4	2	27	11
	PrecStd	204	41	44	64	56	60	35	35	44	57	74	73	53
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	23.6	26.4	28.0	31.5	36.5	37.9	38.2	34.6	30.9	25.7	20.9
		MCWB	15.4	16.9	17.8	20.3	22.3	23.8	23.3	25.6	23.8	21.5	17.3	16.7
	2%	DB	18.0	20.6	24.3	26.4	29.8	34.1	35.5	35.8	32.9	28.8	22.4	19.0
		MCWB	13.1	15.8	17.5	18.6	22.2	24.5	24.6	24.7	23.8	20.4	16.7	15.6
	5%	DB	15.3	18.4	21.9	24.8	28.6	32.6	33.9	34.0	31.4	26.4	20.4	16.2
		MCWB	10.9	13.4	15.3	18.2	21.9	24.2	25.1	24.3	23.2	18.6	15.6	12.7
	10%	DB	12.7	15.8	19.7	23.1	27.0	31.2	32.5	32.2	29.4	24.2	18.1	13.5
		MCWB	8.7	12.0	14.1	17.3	21.1	23.8	25.0	24.2	22.2	17.8	13.6	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	18.4	19.3	21.7	24.4	26.6	27.2	27.1	25.4	23.7	20.2	18.1
		MCDB	19.2	22.7	25.2	26.0	29.3	32.7	32.9	33.6	32.1	29.2	23.8	20.4
	2%	WB	14.2	16.8	17.9	20.4	23.5	25.7	26.5	26.3	24.7	22.2	18.1	16.2
		MCDB	16.4	20.2	23.0	24.9	28.2	31.9	32.5	32.6	31.0	26.8	21.1	18.6
	5%	WB	11.6	14.9	16.4	19.5	22.7	25.2	25.9	25.6	24.0	20.6	16.2	13.4
		MCDB	14.1	17.4	20.2	23.6	27.1	31.2	31.8	31.5	29.9	24.4	19.6	15.7
	10%	WB	8.9	12.1	14.9	18.5	21.8	24.5	25.3	25.0	23.2	19.2	14.1	10.8
		MCDB	12.6	14.8	18.2	22.3	26.2	30.3	31.2	30.4	28.7	23.0	17.3	12.8
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.7	12.7	12.9	11.8	12.0	11.9	12.1	12.8	13.8	13.1	11.2
		MCDBR	15.4	14.8	15.1	14.3	13.2	13.7	13.5	14.5	14.7	15.6	14.7	14.0
		MCWBR	11.4	11.0	9.9	8.4	6.9	6.0	5.0	5.4	6.4	8.3	9.7	10.5
	5% WB	MCDBR	12.2	12.7	13.5	11.1	11.2	11.9	11.7	12.1	13.1	13.3	11.6	11.4
		MCWBR	10.5	10.0	9.4	7.6	6.5	5.7	5.2	5.3	6.0	8.4	9.0	9.7
Clear-Sky Solar Irradiance	taub	0.314	0.322	0.357	0.389	0.430	0.453	0.485	0.474	0.408	0.355	0.325	0.308	
	taud	2.487	2.463	2.399	2.307	2.253	2.208	2.164	2.172	2.346	2.475	2.513	2.521	
	Ebn at Noon	886	920	911	894	858	835	805	806	847	870	867	872	
	Edh at Noon	83	95	111	128	138	144	149	145	116	93	81	76	
All-Sky Solar Radiation	RadAvg	2.43	2.98	4.09	5.17	5.68	6.50	6.22	5.71	4.87	3.71	2.63	2.09	
	RadStd	0.21	0.38	0.31	0.37	0.46	0.44	0.32	0.36	0.37	0.46	0.28	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page